Name: _____

Date: _____

Mr. Rawson • Y8 MYP Physics

2.2 – Newton's 2nd Law

Unit 2: Forces and Motion

Before you watch

A skydiver jumps from a plane. Her speed changes several times before she lands — speeding up, then steady, then slowing down, then steady again. This video is only about three and a half minutes long. Watch it once all the way through, then we go back over it together.

Listen for one pair of words: **balanced** and **unbalanced**.

While you watch

1. Name the **two** forces acting on the skydiver as she falls.

   _____

2. One of those two forces stays the same the whole time. Which one?

   _____

3. The other force changes as she falls. Does it get bigger or smaller as her speed increases?

   _____

Newton's Second Law

> **When the forces on an object are unbalanced, the object accelerates — it speeds up, slows down, or changes direction — in the direction of the *bigger* force. When the forces are balanced, there is no acceleration: the object's speed stays constant.**

Check: "balanced" does not mean "stopped." A skydiver falling at a steady speed has balanced forces *and* is still moving — Newton's First Law again. Newton's Second Law is about what happens the moment the forces *stop* being balanced.

The skydive, in four stages

Fill in the last column yourself, using **speeding up**, **constant speed**, or **slowing down**.

Stage	Forces	What happens to her speed
Just after jumping	gravity much bigger than air resistance	
Falling for a while	air resistance has grown until it equals gravity	
Parachute opens	air resistance suddenly much bigger than gravity	
After the parachute has been open a while	air resistance has shrunk back down until it equals gravity again	

Apply it

1) **State** Newton's Second Law, in your own words.

> **State:** Give a specific name, value or other brief answer without explanation or calculation.

Model answer: When the forces on an object are unbalanced, the object accelerates in the direction of the bigger force. When the forces are balanced, the object's speed stays constant.

2) **Describe** how the two forces on the skydiver compare in the first few seconds after she jumps, and what happens to her speed because of it.

> **Describe:** Give a detailed account or picture of a situation, event, pattern or process.

Model answer: Straight after jumping, gravity is much bigger than air resistance, so the forces are unbalanced. Because the unbalanced force acts downwards, she accelerates downwards — her falling speed increases.

3) **Explain** why the skydiver's falling speed stops increasing even though gravity has not changed.

> **Explain:** Give a detailed account including reasons or causes.

Model answer: Air resistance increases as her falling speed increases. Eventually air resistance grows large enough to equal gravity. Once the two forces are balanced, there is no longer an unbalanced force to accelerate her, so her speed stops changing and stays constant — even though gravity itself is still acting on her the whole time.

Extension

Think further. When the parachute first opens, the skydiver is briefly pushed *upward* in her seat straps, even though she is still moving downward overall. Using Newton's Second Law, suggest why.

The instant the parachute opens, air resistance becomes much bigger than gravity, so the resultant (unbalanced) force now points upward, even though she is still travelling downward. By Newton's Second Law, an unbalanced force causes acceleration in its direction, so she decelerates sharply — which feels, briefly, like being pulled upward in the harness.